HW7 | CTLE simulation and design

Anonymous numerical report

A two-pole, one-zero CTLE was tuned for the supplied 50 cm stripline at 40 Gbps. The center eye height increases from -107.40 mV to 360.61 mV, with a PDA margin of 319.31 mV. All assignment constraints are checked explicitly.

Selected CTLE parameters

Parameter	Selected value	Meaning
f_p1	20.600000 GHz	First pole frequency
f_p2	22.800000 GHz	Second pole frequency
A_DC	-4.910000 dB	DC amplitude gain
A_AC	11.480000 dB	Midband gain parameter
f_z	3.121528 GHz	Derived zero frequency

Transfer-function convention

$H_{eq}(s) = a_{AC}\ \mathrm{omega}_{p2}\ [s + (a_{DC}/a_{AC})\ \mathrm{omega}_{p1}] / [(s + \mathrm{omega}_{p1})(s + \mathrm{omega}_{p2})]$

Here $a_{DC} = 10^{A_DC/20}$, $a_{AC} = 10^{A_AC/20}$, and $\mathrm{omega}_p = 2\ \mathrm{pi}\ f_p$. The zero is $f_z = (a_{DC}/a_{AC})\ f_{p1}$. A_{AC} controls the midband asymptote; the actual maximum is evaluated separately. [1, 2]

Assignment acceptance checks

Constraint	Computed result	Status
1 <= fp1, fp2 <= 40 GHz	20.600, 22.800 GHz	PASS
Maximum CTLE gain <= 6 dB	5.9800 dB	PASS
-20 <= A_DC <= +6 dB	-4.910 dB	PASS
Center eye height >= 300 mV	360.61 mV	PASS

The exact continuous-frequency peak occurs at 21.216523 GHz. The peak check is not restricted to the plotted frequency samples.

The four tuning parameters are the two pole frequencies and the DC / AC amplitude-gain parameters. No independent bound is imposed on A_AC; the complete response must satisfy the 6 dB maximum.

Frequency response and channel model

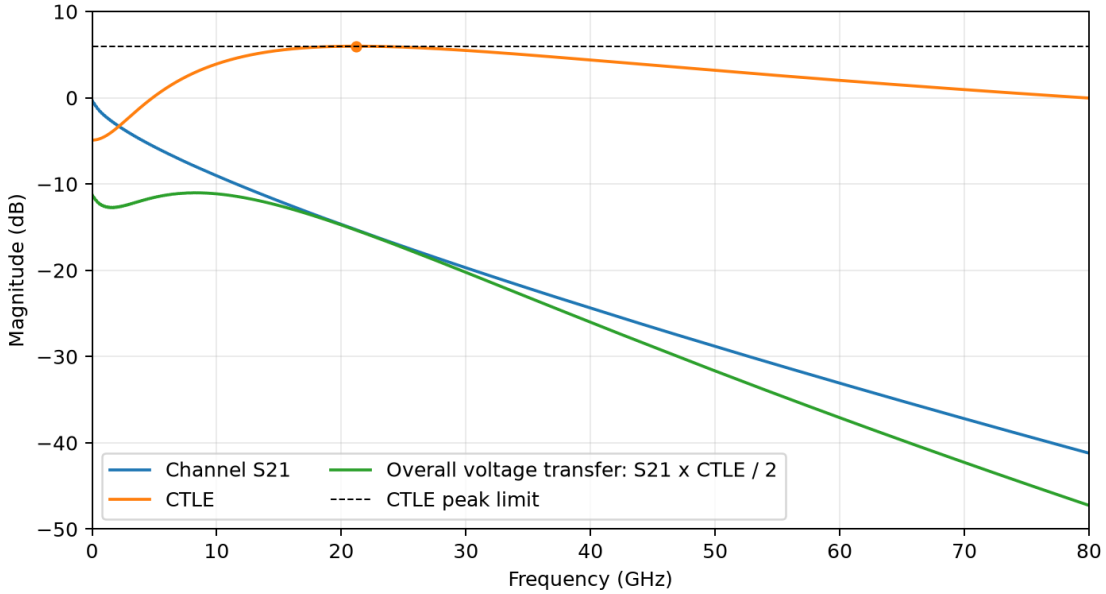


Figure 1. Optimized CTLE magnitude and the supplied channel / combined response. The CTLE compensates the channel's smooth low-pass loss over the useful signal band.

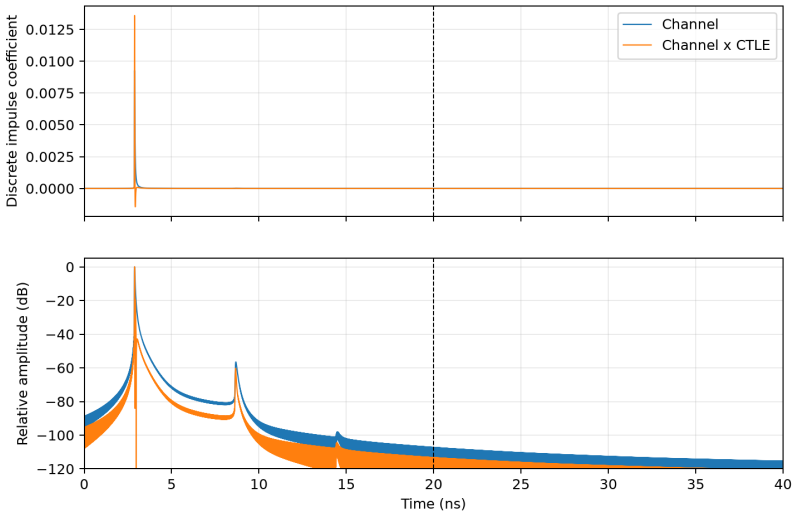


Figure 2. Impulse responses used for convolution. Both paths use the same interpolation, frequency grid, taper and time sampling.

The input file contains 8,001 complex frequency samples with spacing 10 MHz, a maximum frequency of 80 GHz, and reference impedance 50 ohm. Its complex S21 is retained, including phase.

The positive-frequency spectrum is Kaiser-tapered using the actual data cutoff (80 GHz), then zero-padded to the simulation Nyquist frequency before a real inverse FFT. The taper uses alpha = 1; the retained impulse-response window is 20 ns. In particular, no independent hardcoded cutoff is used.

Eye diagrams and center eye height

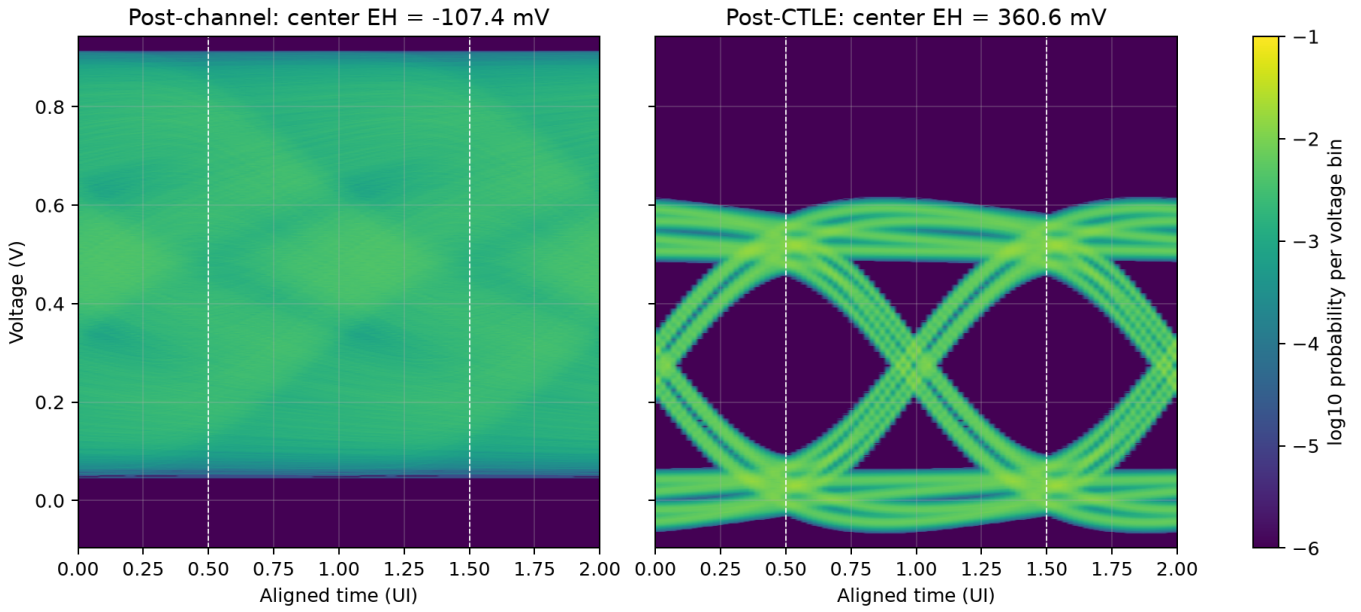


Figure 3. Statistical eyes from the same seeded NRZ sequence. The selected sampling centers are aligned to 0.5 and 1.5 UI in each panel.

Center metric	Post-channel	Post-CTLE
Random-pattern eye height (mV)	-107.40	360.61
PDA eye height (mV)	-198.33	319.31
Minimum conditional one level (mV)	426.35	454.72
Maximum conditional zero level (mV)	533.76	94.12
Selected SBR sample time (ns)	2.9020	2.9005
Sample index within the UI	4	1
Analyzed bits	64,736	64,736

Metric and sampling convention

Each path uses its own fixed sample phase, selected by maximizing the pattern-dependent analysis (PDA) margin around the main cursor. At that phase, random-pattern center EH = min(v | bit = 1) - max(v | bit = 0), over every retained bit. No probability threshold removes rare excursions. A negative separation means the one / zero distributions overlap and the physical eye is closed.

PDA sums the worst-case sampled precursor and postcursor contributions. It checks more bit combinations than the finite random sequence and is reported separately from its observed eye height.

Both paths use the same 65,536 bits (seed 0) at 40 Gbps. UI = 25 ps; the source has 5 ps full 0-to-100% linear ramps, following the supplied sample convention. Each UI has 50 samples (dt = 0.5 ps). The first 800 bits are excluded to remove the startup transient.

Three 1 ns waveform comparisons

The overlays show the received post-channel and post-CTLE voltages in three different 1 ns intervals. Each trace is taken from the same complete simulated record; the input sequence and voltage scale are unchanged.

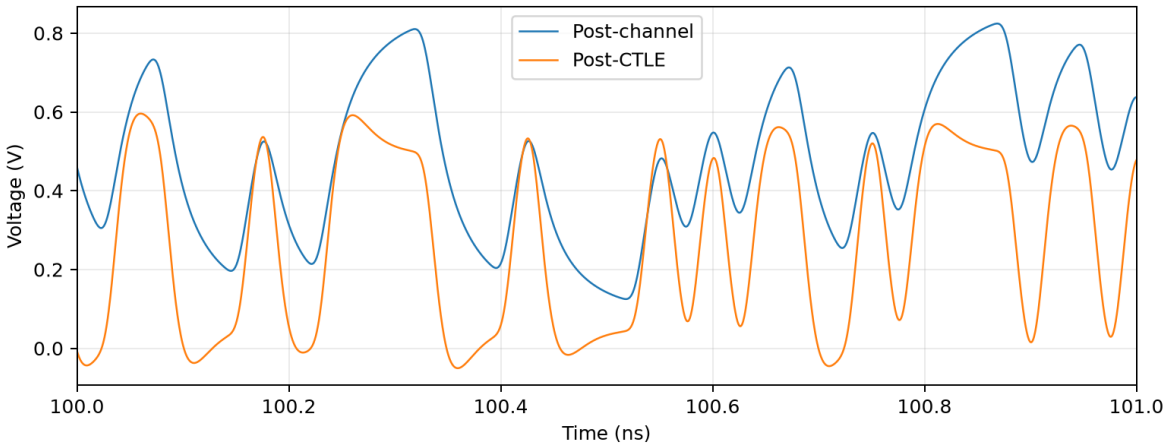


Figure 4. Waveform interval 100-101 ns. Both received paths are plotted together.

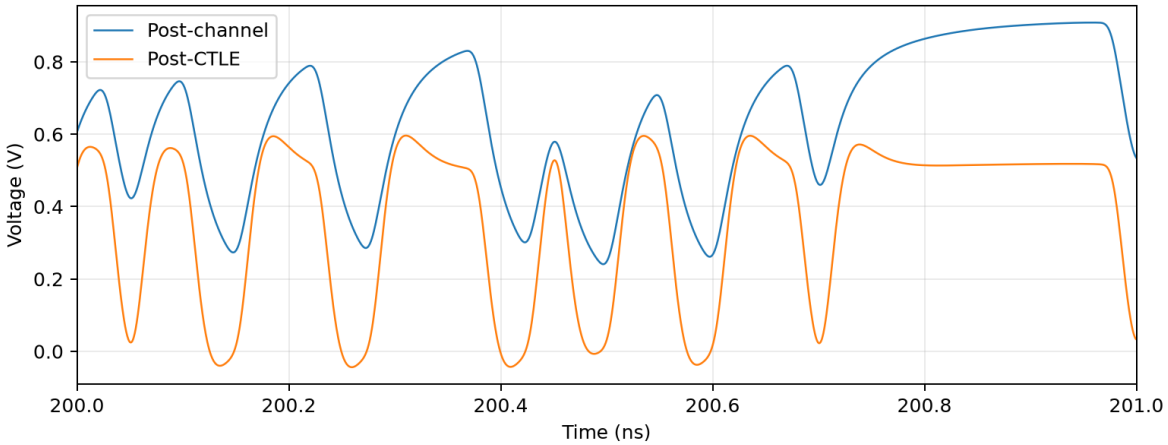


Figure 5. Waveform interval 200-201 ns. Both received paths are plotted together.

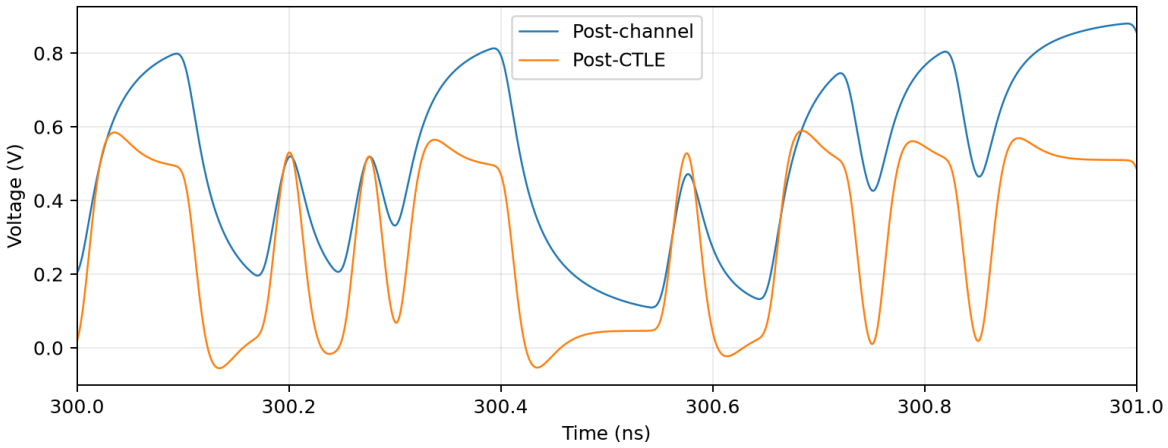


Figure 6. Waveform interval 300-301 ns. Both received paths are plotted together.

The CTLE reduces the long channel tail and separates the received logic levels. It may also produce overshoot or negative undershoot; the center eye-height calculation includes their contribution at the selected sample phase.

Reproducibility and verification

Amplitude and convolution accounting

The source is a 2 Vpp unipolar NRZ waveform with 5 ps linear transitions. Under the assignment's matched-source assumption, the post-channel transfer is H_ch = S21_ch / 2 and the post-CTLE transfer is H_total = S21_ch S21_CTLE / 2. The matching divider 0.5 is applied exactly once.

Sectional convolution is overlap-add with a retained output tail. The inverse FFT produces a discrete impulse response, so no extra dt factor is introduced. The full-period sample sum equals the DC transfer; truncation slightly changes that sum and is not renormalized. A single isolated bit with the same rising and falling ramps supplies the PDA response.

Independent numerical checks

Independent reader and gain checks. All 12 unit/format combinations reproduce a defined two-port matrix. A dense 12-decade frequency sweep and a separately derived peak calculation confirm the global CTLE gain limit.

Impulse and physical normalization. The real IFFT matches a full Hermitian IFFT and preserves the DC kernel sum. An identity transfer maps the 2 V source high level to 1 V with exactly one matched-source division by two.

Direct versus sectional convolution. Overlap-add matches direct linear convolution for a 137-bit pattern, including its partial final block and final falling ramp; transition endpoints independently match time-knot interpolation.

Eye alignment and sensitivity. Bit-conditioned random EH and all-tap PDA use y[k*samples_per_ui+cursor] against bit k and remain distinct measures. An independent continuous-pulse spectrum, 0.25 ps grid and full 100 ns impulse period retain more than 300 mV worst-case margin.

Tail and time-step robustness. Extending the kernel from 20 to the full 100 ns gives PDA EH = 317.824 mV. Halving dt to 0.25 ps gives 317.717 mV with the full kernel, a change of -0.107 mV. Both exceed 300 mV.

Interpretation and limits

This linear, frequency-domain, noise-free model retains channel phase and delay, but omits noise, clock jitter, nonlinearity and saturation. CTLE can amplify high-frequency noise and does not generally remove reflection notches. Finite-pattern EH and finite-period PDA are not BER predictions or validation of an analog implementation.

The optional ADS ChannelSim exercise was not performed. Figures and claims come from the accompanying Python calculation and independent checks.

References and provenance

[1] Assignment brief, *Homework: CTLE simulation and design*, instructions/HW7_CTLE.pdf, pp. 1-3. Supplied channel and sample code are stored alongside it.

[2] Course lecture, *Continuous Time Linear Equalizer (CTLE)*, pp. 6-9 and 11-14. Transfer-function conventions, circuit interpretation and over-equalization.

Channel SHA-256: 564560062e750405e89cadbb8be73cc7a40ad3188d19383ebbfe1e22723f5efc